

Muhammad Haroon Khan

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EDUCATION

Ph.D. Physics	Aug 2022 - Present	University of Houston, TX
M.S. Physics	Aug 2022 – Dec 2025	University of Houston, TX
M. Phil Physics	Feb 2020 - Apr 2022	Quaid-i-Azam University, Islamabad
B.S. Physics	Sep 2015 - Jan 2020	Quaid-i-Azam University, Islamabad

HONORS & AWARDS

❖ Graduate Research Assistantship	2022 - 2027	Higher Education Commission of Pakistan
❖ TcSUH Graduate Student Travel Support Award	2026 - 2027	Texas Center for Superconductivity at University of Houston
❖ Winner of the Best Presentation Award in Experimental Condensed Matter Session	2025	University of Houston Research Day
❖ Departmental Fellowship for outstanding grades in M. Phil coursework	2022	Quaid-i-Azam University, Islamabad

LEADERSHIP AND VOLUNTEERING EXPERIENCE

❖ TEX-E Fellow	2026 - 2027	TEX-E
❖ Vice Chair of Mentorship Committee	Fall 2025 - Present	UH Energy Coalition
❖ Assistant Session Chair	May 17–21, 2026	Internation Thermoelectric Conference 2026, Houston
❖ NSM Grad. Student Council Representative	2024 - 2025	Department of Physics at the University of Houston
❖ Vice Chair of Sustainability Committee	Spring 2025	UH Energy Coalition
❖ Judge in Physics and Astronomy category at Science & Engineering Fair Houston	2022 - 2026	SEFH, Houston

WORK EXPERIENCE

Graduate Student Researcher	Aug 2023 – Present	University of Houston, TX
❖ Supervisor: Dr. Shuo Chen		
❖ Investigating high-performance Mg_3Sb_2 -based thermoelectric materials for low-grade industrial waste heat recovery through transition-metal doping, defect engineering, phase composition optimization and conversion efficiency device fabrication.		

- ❖ Characterizing interface defects and microstructural properties of high-temperature superconducting single crystals via advanced TEM techniques, including electron diffraction and energy-dispersive X-ray spectroscopy (EDS).
- ❖ Developing novel, low-cost electrocatalysts for green hydrogen production through electrochemical water-splitting, correlating structural properties via XRD and TEM with catalytic performance and stability in industrial-scale water electrolyzer systems.

Teaching Assistant

Fall 2024

University of Houston, TX

- ❖ Teaching Assistant in College Physics Lab 1
- ❖ Demonstrated students with their weekly lab experiments, conducted weekly office hours, class lab report grading, as well as exams grading.

RESEARCH PUBLICATIONS

Published

- ❖ **Muhammad Haroon Khan**, Rui Shu, Shuo Chen, "Phase Segregation in High Mn-doped Mg_3SbBi Based Thermoelectrics." [ACS Applied Energy Materials](#) (2025)
- ❖ Fiza Mumtaz, **Muhammad Haroon Khan**, G. Hassnain Jaffari. "Correlation between the composition, phase, band structure, ferroelectric and leakage responses of $\text{Bi}_{1-x}\text{Ba}_x\text{Fe}_{1-y}\text{Ti}_y\text{O}_3$ thin films." [Thin Solid Films](#) (2022).
- ❖ Publications can be accessed on my [Google Scholar Profile](#).

Manuscripts in Preparation

- ❖ **Muhammad Haroon Khan**, L. Deng, S. Chen *et al.*, "Transmission Electron Microscopy Study of High T_c Hg-1223 Superconducting Materials." Manuscript in preparation.
- ❖ P. Byaruhanga, **Muhammad Haroon Khan**, S. Chen *et al.*, "Synergistic Sulfur-Selenium Co-Doping of NiFe LDH Enhanced Oxygen Evolution." Manuscript in preparation.

PRESENTATIONS AND WORKSHOPS

- ❖ **Muhammad Haroon Khan**, Shuo Chen, " Role of Phase Purity in $\text{Mg}_{3-x}\text{Mn}_x(\text{Sb,Bi})_2$ Thermoelectric Materials ", The 42nd International Conference on Thermoelectrics 2026, Houston, TX.
- ❖ **Muhammad Haroon Khan**, Rui Shu, Shuo Chen " Phase segregation in high Mn-doped Mg_3SbBi thermoelectric materials". [American Physical Society](#) & TSAAPT Joint Spring Meeting (2025).
- ❖ Paul Byaruhanga, **Muhammad Haroon Khan**, Shuo Chen "Investigating Effect of Dual Doping S and Se on NiFe Layered Double Hydroxide Catalyst for Oxygen Evolution Reaction". [Electrochemical Society Meeting](#) (2025).
- ❖ Tech boot camp on robophysics Summer 2024
- ❖ IPOLS Scientific Programming Summer 2024
- ❖ International School on Rietveld Refinement & Pair Distribution ([ISBARM-22](#)) Summer 2022

Georgia Institute of Technology, GA
Emory University, GA

National Centre for Physics, Pakistan